Multiplying and Dividing Fractions

Name _____ Date _____ No calculator Version 1

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KEY WORDS numerator *(taurunga)* · denominator *(tauraro)* · product *(otinga whakarea)* · reciprocal

A A fraction of an amount ●○○

Remember: Divide by the bottom, then multiply by the top: $\frac{3}{4}$ of 20 is $20 \div 4 \times 3 = 15$.

1. Work out.

a) $\frac{1}{4}$ of 28 = _____ **b)** $\frac{2}{3}$ of 18 = _____ **c)** $\frac{3}{5}$ of 35 = _____ **d)** $\frac{5}{6}$ of 42 = _____

e) $\frac{3}{8}$ of 56 = _____ **f)** $\frac{7}{10}$ of 90 = _____ **g)** $\frac{4}{9}$ of 63 = _____ **h)** $\frac{2}{7}$ of 84 = _____

i) $\frac{3}{4}$ of 120 m = _____ **j)** $\frac{5}{8}$ of 240 g = _____ **k)** $\frac{2}{5}$ of 65 km = _____ **l)** $\frac{11}{12}$ of 60 min = _____

B A fraction times a whole number ●○○

Remember: Multiply the top by the whole number. Write an improper fraction as a mixed number.

2. Work out. Simplify your answer where you can.

a) $\frac{1}{3} \times 2$ = _____ **b)** $\frac{2}{5} \times 2$ = _____ **c)** $\frac{3}{4} \times 3$ = _____ **d)** $\frac{2}{7} \times 5$ = _____

e) $\frac{5}{6} \times 4$ = _____ **f)** $\frac{3}{8} \times 6$ = _____ **g)** $\frac{4}{9} \times 3$ = _____ **h)** $\frac{7}{10} \times 5$ = _____

C Multiplying fractions ●●○

Remember: Multiply the tops together and multiply the bottoms together. Then simplify.

3. Work out. Simplify your answer where you can.

a) $\frac{1}{2} \times \frac{1}{3}$ **b)** $\frac{2}{3} \times \frac{1}{5}$ **c)** $\frac{3}{4} \times \frac{1}{2}$

d) $\frac{2}{5} \times \frac{3}{7}$ **e)** $\frac{3}{4} \times \frac{2}{3}$ **f)** $\frac{5}{6} \times \frac{3}{10}$

g) $\frac{4}{9} \times \frac{3}{8}$ **h)** $\frac{7}{8} \times \frac{4}{5}$ **i)** $\frac{5}{12} \times \frac{3}{5}$

j) $\frac{9}{10} \times \frac{5}{6}$ **k)** $\frac{7}{3} \times \frac{6}{5}$ **l)** $\frac{8}{3} \times \frac{9}{4}$

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D Reciprocals ●●○

> **Remember:** To find the reciprocal, turn the fraction upside down. A number times its reciprocal is 1.

4. Write the reciprocal of each number.

a) $\frac{3}{4}$ _____ **b)** $\frac{2}{5}$ _____ **c)** 5 _____ **d)** $\frac{1}{6}$ _____

e) $\frac{7}{3}$ _____ **f)** 12 _____ **g)** $2\frac{1}{2}$ _____ **h)** $1\frac{2}{3}$ _____

E Dividing fractions ●●●

> **Remember:** To divide by a fraction, multiply by its reciprocal: $\frac{2}{3} \div \frac{4}{5} = \frac{2}{3} \times \frac{5}{4}$.

5. Work out. Simplify your answer where you can.

a) $\frac{1}{2} \div \frac{1}{4}$ **b)** $\frac{2}{3} \div \frac{1}{3}$ **c)** $\frac{3}{4} \div \frac{1}{2}$

d) $\frac{2}{5} \div \frac{3}{10}$ **e)** $\frac{5}{6} \div \frac{2}{3}$ **f)** $\frac{3}{8} \div \frac{3}{4}$

g) $\frac{4}{7} \div \frac{2}{5}$ **h)** $\frac{7}{9} \div \frac{2}{3}$ **i)** $\frac{5}{8} \div \frac{15}{16}$

j) $\frac{3}{10} \div \frac{6}{5}$ **k)** $\frac{9}{4} \div \frac{3}{8}$ **l)** $\frac{7}{12} \div \frac{14}{3}$

F Problems ●●●

6. One batch of muffins needs $\frac{3}{4}$ of a cup of milk. How much milk is needed for 6 batches?

7. A 6 m rope is cut into pieces that are each $\frac{3}{8}$ m long. How many pieces are there?

8. A tank holds 45 litres when it is full. It is $\frac{3}{5}$ full. How many litres are in the tank?

9. There are 24 students in a class. $\frac{2}{3}$ of them walk to school, and $\frac{1}{4}$ of the walkers live in the same street. How many students is that?

★ Extension

10. Change each mixed number to an improper fraction first.

a) $1\frac{1}{2} \times 2\frac{2}{3}$ **b)** $3\frac{3}{4} \div 1\frac{1}{4}$

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Multiplying and Dividing Fractions

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS numerator *(taurunga)* · denominator *(tauraro)* · product *(otinga whakarea)* · reciprocal

A A fraction of an amount ●○○

Remember: Divide by the bottom, then multiply by the top: $\frac{3}{4}$ of 20 is $20 \div 4 \times 3 = 15$.

1. Work out.

a) $\frac{1}{5}$ of 35 = _____ **b)** $\frac{2}{3}$ of 24 = _____ **c)** $\frac{3}{4}$ of 36 = _____ **d)** $\frac{5}{6}$ of 54 = _____

e) $\frac{3}{8}$ of 72 = _____ **f)** $\frac{7}{10}$ of 60 = _____ **g)** $\frac{4}{9}$ of 81 = _____ **h)** $\frac{2}{7}$ of 63 = _____

i) $\frac{3}{5}$ of 150 m = _____ **j)** $\frac{5}{8}$ of 320 g = _____ **k)** $\frac{2}{3}$ of 75 km = _____ **l)** $\frac{7}{12}$ of 60 min = _____

B A fraction times a whole number ●○○

Remember: Multiply the top by the whole number. Write an improper fraction as a mixed number.

2. Work out. Simplify your answer where you can.

a) $\frac{1}{4} \times 3$ = _____ **b)** $\frac{2}{7} \times 3$ = _____ **c)** $\frac{2}{3} \times 4$ = _____ **d)** $\frac{3}{5} \times 4$ = _____

e) $\frac{5}{8} \times 4$ = _____ **f)** $\frac{5}{6} \times 3$ = _____ **g)** $\frac{2}{9} \times 6$ = _____ **h)** $\frac{3}{10} \times 5$ = _____

C Multiplying fractions ●●○

Remember: Multiply the tops together and multiply the bottoms together. Then simplify.

3. Work out. Simplify your answer where you can.

a) $\frac{1}{2} \times \frac{1}{5}$ **b)** $\frac{3}{4} \times \frac{1}{3}$ **c)** $\frac{2}{3} \times \frac{1}{4}$

d) $\frac{3}{5} \times \frac{2}{7}$ **e)** $\frac{2}{3} \times \frac{3}{8}$ **f)** $\frac{5}{6} \times \frac{2}{5}$

g) $\frac{3}{10} \times \frac{5}{9}$ **h)** $\frac{5}{8} \times \frac{4}{7}$ **i)** $\frac{7}{12} \times \frac{3}{7}$

j) $\frac{4}{5} \times \frac{5}{6}$ **k)** $\frac{5}{2} \times \frac{4}{3}$ **l)** $\frac{9}{4} \times \frac{8}{3}$

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D Reciprocals ●●○

Remember: To find the reciprocal, turn the fraction upside down. A number times its reciprocal is 1.

4. Write the reciprocal of each number.

a) $\frac{2}{3}$ _____ **b)** $\frac{3}{8}$ _____ **c)** 7 _____ **d)** $\frac{1}{9}$ _____

e) $\frac{9}{4}$ _____ **f)** 15 _____ **g)** $3\frac{1}{2}$ _____ **h)** $1\frac{3}{4}$ _____

E Dividing fractions ●●●

Remember: To divide by a fraction, multiply by its reciprocal: $\frac{2}{3} \div \frac{4}{5} = \frac{2}{3} \times \frac{5}{4}$.

5. Work out. Simplify your answer where you can.

a) $\frac{1}{2} \div \frac{1}{6}$ **b)** $\frac{3}{4} \div \frac{1}{4}$ **c)** $\frac{2}{3} \div \frac{1}{2}$

d) $\frac{3}{5} \div \frac{9}{10}$ **e)** $\frac{5}{6} \div \frac{1}{3}$ **f)** $\frac{5}{8} \div \frac{3}{4}$

g) $\frac{3}{7} \div \frac{2}{5}$ **h)** $\frac{8}{9} \div \frac{2}{3}$ **i)** $\frac{3}{8} \div \frac{9}{16}$

j) $\frac{7}{10} \div \frac{14}{5}$ **k)** $\frac{8}{3} \div \frac{4}{9}$ **l)** $\frac{5}{12} \div \frac{10}{3}$

F Problems ●●●

6. One batch of muffins needs $\frac{2}{3}$ of a cup of milk. How much milk is needed for 5 batches?

7. A 9 m rope is cut into pieces that are each $\frac{3}{4}$ m long. How many pieces are there?

8. A tank holds 56 litres when it is full. It is $\frac{5}{8}$ full. How many litres are in the tank?

9. There are 30 students in a class. $\frac{3}{5}$ of them walk to school, and $\frac{1}{3}$ of the walkers live in the same street. How many students is that?

★ Extension

10. Change each mixed number to an improper fraction first.

a) $2\frac{1}{2} \times 1\frac{3}{5}$ **b)** $4\frac{1}{2} \div 1\frac{1}{2}$

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